

UCS321:AI FOR ENGINEERS

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2	0	2	3.0

Course Objectives: Develop foundational AI knowledge while equipping participants with practical problem-solving skills using data relevant to engineering applications.

Syllabus

1. Introduction to AI for Engineers

- *Definitions:* Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Reinforcement Learning (RL)
- Understanding AI's role in engineering decision-making
- Illustrative examples from engineering fields (e.g., predictive maintenance, process control)

2. Data and Preprocessing

- Types of data: sensor data, image data, text data, and time-series data
- Techniques for cleaning, normalization, and handling missing values
- Introduction to feature selection methods

3. Supervised Learning

- Core algorithms: Linear Regression, Logistic Regression, Decision Trees, Random Forest
- Applications: Predicting equipment failure or component lifespan

4. Unsupervised Learning

- Clustering techniques: K-Means, DBSCAN
- Dimensionality reduction methods: Principal Component Analysis (PCA), t-SNE
- Applications: Process classification and anomaly detection

5. Neural Networks & Deep Learning (Intro)

- Fundamentals of Neural Networks: Multilayer Perceptron (MLP), Convolutional Neural Networks (CNN)
- Use cases: Image recognition (e.g., detecting cracks in structures)

6. Time-Series Analysis

- Forecasting techniques: ARIMA and introductory Long Short-Term Memory (LSTM) models
- Applications: Load prediction and vibration analysis

7. Toolchain for Engineers

- Programming tools: Python, Jupyter Notebook
- Libraries: pandas, scikit-learn, TensorFlow/Keras
- Version control: Using Git effectively

8. Mini Project

- Choose an engineering-specific dataset
- Apply preprocessing, modeling, and evaluation techniques

Lab sessions:

1. Build a regression model to predict temperature rise in a motor
2. Use K-Means clustering to group materials based on properties
3. Create a neural network to detect faulty circuits

Case Studies

1. Siemens: AI applications in industrial automation
2. General Electric (GE): AI-driven turbine health monitoring
3. Honeywell: Predictive maintenance in smart factories

Text Books:

1. E. Alpaydin, Introduction to Machine Learning, 4th ed. MIT Press, 2020.
2. Goodfellow L., Bengio Y. and Courville A., Deep Learning, MIT Press (2016).
3. Haykin S., Neural Network and Machine Learning, Prentice Hall Pearson (2009), 3rd ed.

Reference Books:

1. Patterson J. and Gibson A., Deep Learning: A Practitioner's Approach, O'Reilly (2017), 1st ed.
2. Introduction to Machine Learning with Python, Andreas C. Muller and Sarah Guido, O'Reilly.
3. Machine Learning, The art and science of algorithms that make sense of data, Peter Flach, Cambridge University Press

Evaluation Scheme	
Evaluation Elements	Weightage %
Quizzes	20%
Lab Assignments	20%
Mini Project	40%
Viva/Presentation	20%